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Construction of an explainable machine learning model based on SHAP for predicting cardiovascular mortality risk among American cancer survivors

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Abstract

Background Cardiovascular disease (CVD) continues to be a leading cause of death globally. Cancer survivors experience a significantly increased risk of cardiovascular mortality owing to shared risk factors with CVD; however, dedicated predictive tools for this specific population are currently lacking.

Methods Using data from 3622 cancer survivors in the National Health and Nutrition Examination Survey (NHANES), we constructed a predictive model for cardiovascular mortality risk by integrating five machine learning algorithms: Random Survival Forest (RSF), Gradient Boosting Machine (GBM), LASSO-penalized Cox regression (LASSO + Cox), CoxBoost, and Survival Support Vector Machine (SVM). The SHapley Additive exPlanations (SHAP) interpretability framework was employed to enhance model explainability. Predictors were selected via univariate and multivariate Cox regression analyses, and a nomogram was derived from the results of LASSO regression. To foster clinical translation, the optimal model was implemented as an openly accessible, interactive web-based tool named "CardioRisk-CA" (Cardiovascular Risk Calculator for Cancer Survivors). External validation was conducted using the Surveillance, Epidemiology, and End Results (SEER) database to assess the generalizability of the model.

Results The GBM demonstrated the highest predictive performance in the validation dataset (Area Under the receiver operating characteristic Curve (AUC) = 0.935, Harrell's concordance index (C-index) = 0.914). Key predictive variables included age, C-reactive protein (CRP), neutrophil-to-lymphocyte ratio (NLR), history of stroke, coronary heart disease, heart failure, sex, and marital status. SHAP analysis identified interaction effects between age and stroke, coronary heart disease, and heart failure. The nomogram achieved an AUC of 0.868, indicating strong clinical utility. Kaplan–Meier analysis revealed significant survival differences between high-risk and low-risk groups ($P < 0.001$). The web-based tool generates interpretable, individualized risk assessments for clinicians and provides researchers to perform external validation with their own datasets. In external validation using the SEER cohort, the GBM model continued to show robust discriminative capability (C-index = 0.815, AUC = 0.866).

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Conclusion The developed machine learning model predicts cardiovascular mortality risk in cancer survivors with SHAP-based interpretability. Deployed as an accessible web-based tool, it fosters individualized risk management and facilitates multicenter validation.

Keywords Cancer survivors, Cardiovascular mortality, Machine learning, Predictive model, SHAP, NHANES, SEER

1 Introduction

Advances in cancer diagnosis and treatment have significantly increased the population of cancer survivors. Estimates from the American Cancer Society indicate a steady decline in cancer mortality since 1991, resulting in an overall reduction of 33% and approximately 3.8 million fewer cancer-related deaths [1]. Despite this progress, life expectancy among cancer patients remains shorter than that of the general population. Cardiovascular disease (CVD) and cancer are among the leading causes of death globally. Although frequently viewed as distinct conditions, they share multiple similarities and interact pathophysiologically [2]. Due to several common risk factors, many cancer patients, particularly older adults, experience both conditions simultaneously [3–5]. Epidemiological data consistently identify CVD as the primary non-cancer cause of mortality among cancer survivors [6].

The association between cancer and CVD has been extensively documented. A UK population-based cohort study utilizing linked electronic health records demonstrated that survivors of most site-specific cancers have an elevated medium- to long-term risk of CVD compared with the general population [7]. Shared modifiable risk factors, including obesity, smoking, alcohol consumption, and diabetes, contribute to common biological mechanisms such as chronic inflammation and oxidative stress. Non-modifiable factors such as age, sex, and race also influence disease incidence [2]. Therefore, comprehensively assessing these factors is critical for stratifying cardiovascular mortality risk among cancer survivors. In this context, leveraging large-scale cohort data to develop and validate machine learning models has become a well-established strategy for identifying risk and managing complex diseases [8, 9].

Recent advancements in feature selection for high-dimensional clinical data signify a shift from a narrow focus on statistical significance toward a holistic framework that values clinical interpretability, stability, and expert knowledge. This paradigm is exemplified by the multi-step framework of Wang et al., which integrates data-driven screening with expert validation and stability analysis to ensure selected features are both predictive and clinically meaningful [10]. Guided by this paradigm, we designed a comprehensive feature selection workflow. First, potential predictors were screened using univariate and multivariate Cox regression. Following this, their stability was rigorously evaluated via cross-validation within multiple machine learning algorithms. To conclude, SHAP interpretability analysis—interpreted with clinical expertise—was applied to validate and contextualize the final set of features. This multifaceted strategy not only optimizes predictive accuracy but also directly addresses the opacity inherent in complex models, thereby enhancing their transparency and trustworthiness for real-world clinical application. To strengthen the generalizability of predictive models, we also conducted external validation using the Surveillance, Epidemiology, and End Results (SEER) database, a large population-based cancer registry.

To facilitate translation into practice, a web-based tool hosting the optimal model was developed. This platform not only generates clinically actionable, interpretable risk profiles for individual patients but also supports researchers to conduct external validation studies with their own datasets. By bridging the critical "last mile" between research and clinical practice, this tool serves as a tangible, real-time decision-support system designed to enhance cardiovascular risk management for cancer survivors.

2 Methods

2.1 Data source and study population

We conducted an observational cohort study using data obtained from seven consecutive cycles (2005–2018) of the National Health and Nutrition Examination Survey (NHANES), a nationally representative survey administered by the Centers for Disease Control and Prevention (CDC). NHANES employs a complex, multistage, probability sampling design with oversampling of specific demographic subgroups to ensure robust national representation. The survey collects comprehensive data on health and nutrition through interviews, physical examinations, and laboratory tests. Owing to its national representativeness, comprehensive variables, and long-term follow-up data, the NHANES database has been widely utilized for developing machine learning-based risk prediction models in cancer and cardiovascular research [9, 11]. From this database, we identified participants aged ≥ 20 years who reported a history of cancer (cancer survivors) and had available linkage to CVD mortality data. After excluding individuals with missing data on key covariates, the final analytic cohort comprised 3622 cancer survivors. A flowchart detailing the participant selection process is provided in Supplementary Fig. 1. The NHANES protocol was approved by the National Center for Health Statistics (NCHS) Research Ethics Review Board, and all participants provided written informed consent.

2.1.1 Cancer survivor definition

Cancer survivor status was determined by an affirmative response to the survey question: "Has a doctor or other health professional ever told you that you had cancer or a malignancy of any kind?"

2.1.2 Cardiovascular mortality endpoint

The primary endpoint was cardiovascular mortality, defined according to the International Classification of Diseases, Tenth Revision (ICD-10) codes I00–I09, I11, I13, I20–I51, or I60–I69 (data available through December 2019).

2.1.3 Covariates

Baseline covariates were extracted from questionnaire and examination data. Demographic characteristics included age, sex, race (categorized as Mexican American, Other Hispanic, Non-Hispanic White, Non-Hispanic Black, or Other Race), educational attainment (< high school, high school graduate, > high school), and marital status (married/living with partner or widowed/divorced/separated/never married). Socioeconomic status was assessed using the poverty-to-income ratio (PIR) and classified as low (PIR ≤ 1.30), middle (PIR = 1.31–3.50), or high income (PIR > 3.5) [12].

2.1.4 Anthropometric and behavioral factors

Body mass index (BMI) was calculated from measured height and weight. Smoking status was classified as never smoker (< 100 lifetime cigarettes), former smoker (≥ 100 lifetime cigarettes, not current), or current smoker (≥ 100 lifetime cigarettes, currently smoking). Alcohol consumption was categorized based on self-reported intake as non-drinker (< 12 drinks/year) or drinker (≥ 12 drinks/year).

2.1.5 Comorbidities

Diabetes mellitus was defined as self-reported physician diagnosis or HbA1c ≥ 6.5%. A composite history of heart disease was established through self-reported physician diagnosis of coronary heart disease, angina, heart failure, or other heart conditions. History of stroke, hypertension, and hypercholesterolemia were similarly based on self-reported physician diagnoses.

2.1.6 Depression and sleep disturbance

Depressive symptoms were assessed using the 9-item Patient Health Questionnaire (PHQ-9), which evaluates symptom severity over the preceding 2 weeks and has validated psychometric properties in cancer populations [13, 14]. Total PHQ-9 scores were categorized as: 0–4 (none), 5–9 (mild), and ≥ 10 (moderate-to-severe depression) [14].

Sleep disturbance was evaluated with the question: "Over the last 2 weeks, how often have you been bothered by trouble falling asleep, staying asleep, or sleeping too much?" Response options included "not at all," "several days," "more than half the days," and "nearly every day." Participants responding "not at all" were classified as having no sleep disturbance, while all other responses indicated sleep disturbance [15].

2.1.7 Laboratory measurements

The NLR was calculated as the ratio of absolute neutrophil count to absolute lymphocyte count. Both counts (10^9 cells/L) were determined using a Beckman Coulter MAXM analyzer.

CRP levels were measured using different platforms across NHANES cycles. From 2005 to 2010, CRP was quantified using a Behring nephelometer with latex-enhanced immunoassay, having a lower limit of detection (LLOD) of 0.02 mg/dL. For the 2015–2018 cycles, a Beckman UniCel DxC 600 analyzer was employed, with LLODs of 0.011 mg/dL (2015–2016) and 0.015 mg/dL (2017–2018). CRP values below the assay-specific LLOD were imputed as LLOD divided by √2.

2.2 Statistical methods

Missing covariate data were imputed separately within the training and validation sets using the Random Forest algorithm. Predictors identified as significant ($P < 0.05$) in univariate Cox regression were evaluated in a multivariate Cox model; variables remaining significant were retained as candidate features for the machine learning models. The dataset was randomly partitioned into a training set (70%) for model development and a validation set (30%) for performance evaluation, a standard practice in machine learning model development [8, 16]. Five survival-based machine learning algorithms were implemented: RSF, GBM, LASSO+ Cox, CoxBoost, and Survival SVM. Hyperparameters were optimized via cross-validation on the training set to avoid overfitting. Model

performance was evaluated using the area under the receiver operating characteristic curve (AUC) and Harrell's concordance index (C-index). Bootstrap resampling (1000 iterations) was used to estimate 95% confidence intervals for these metrics. Survival differences between risk groups were compared using Kaplan–Meier curves and the log-rank test. The SHAP framework was applied to interpret the optimal model. SHAP quantifies the marginal contribution of each feature to individual predictions, providing both global interpretation (feature importance, interactions) and local explanations of specific cases. Thus, SHAP addresses the "black-box" limitation of complex ML models.

All analyses were conducted using R statistical software (version 4.4.3) with a two-sided alpha level of 0.05.

2.3 Development of the clinical prediction web-based tool

To enhance translational impact, we developed "CardioRisk-CA"—an R Shiny-based web tool that publicly hosts the optimal model and its interpretability framework, enabling intuitive risk assessment and validation. The model pipeline was saved (saveRDS) and integrated into an R Shiny application deployed on the shinyapps.io cloud platform, featuring a responsive front-end for multi-device access. To ensure computational efficiency for real-time interaction, SHAP values are calculated using an approximation method based on training set subsampling. This open-access tool facilitates the translation of research findings into clinical practice and supports external validation.

This web-based tool allows users to input key clinical variables. The system then calculates and displays a cardiovascular mortality risk score with corresponding stratification (high/low), and generates individualized SHAP waterfall plots and dependence plots to visually interpret each feature's contribution. Additionally, this tool enables researchers to perform batch prediction and validation.

2.4 External validation using the SEER database

To evaluate the generalizability of our model in an independent, large-scale clinical setting, we conducted external validation using the SEER database (National Cancer Institute). SEER is a population-based registry covering approximately 34.6% of the U.S. population, providing detailed information on cancer incidence, first-course treatment, and survival outcomes. Data were obtained from the SEER Research Data (Incidence—17 Registries, Nov 2023 Sub, 2000–2021). The study included adult patients (aged $\geq$ 20 years) diagnosed with breast, lung, prostate, or colorectal cancer who had a known cancer stage, and excluded those with missing treatment information or unknown vital status.

The previously developed and optimized GBM model (trained on NHANES data) was directly applied to this independent SEER cohort without retraining. Input features included variables common to both NHANES and SEER (age, sex, marital status) and cancer-specific variables available only in SEER (primary cancer site, SEER historic stage, treatment modality). The model generated cardiovascular death risk scores for each patient in the SEER cohort. Discriminative performance (C-index) was evaluated, and survival analyses were conducted based on predicted risk categories. SHAP analysis was subsequently performed on SEER cohort predictions to provide global feature importance within this external validation setting.

3 Results

3.1 Baseline characteristics

Baseline characteristics of participants are summarized in Table 1. All covariates were balanced between the training and validation groups (all $P > 0.05$), confirming the appropriateness of the random split (Supplementary Table 1).

Univariate and multivariate Cox proportional hazards regression analyses were conducted using demographic and clinical covariates. Age, sex, marital status, history of heart failure, coronary heart disease, stroke, NLR, and CRP were identified as independent predictors of cardiovascular mortality among cancer survivors (Supplementary Table 2).

3.2 Model performance evaluation

3.2.1 Discriminative performance (AUC)

The discriminative abilities (AUC values) of the five machine learning models are summarized in Table 2. In the training set, the RSF model achieved the highest AUC (0.989), followed by the GBM model (AUC = 0.927). However, in the independent validation set, the GBM model demonstrated superior generalizability (AUC = 0.935), outperforming the RSF model (AUC = 0.831). Thus, the GBM model was selected as optimal due to its robust performance and generalizability. Corresponding ROC curves for all models are presented in Fig. 1.

3.2.2 Model discrimination: concordance index

The discriminative performance of each model, measured by Harrell's concordance index (C-index) (Table 3). In the training cohort, the RSF model achieved the highest C-index (0.986), followed by the GBM model (0.919). In the independent validation cohort, the GBM model exhibited superior generalizability (C-index = 0.914). The LASSO-penalized Cox (LASSO+Cox) and CoxBoost models showed similar performance (C-index = 0.821). Due to its consistent and superior validation performance, the GBM model was selected as the final optimal model. The time-dependent C-index curves for all five algorithms are presented in Fig. 2.

3.2.3 Benchmarking the GBM model against traditional Cox regression

To assess the incremental value of the machine learning approach, the optimal GBM model was directly compared with a traditional Cox proportional hazards model. Both models utilized an identical set of eight predictor variables. In the independent validation set, the Cox model had an area under the curve (AUC) of 0.840 and a C-index of 0.827. The GBM model displayed superior discrimination, achieving an AUC of 0.935 and a C-index of 0.914 (Supplementary Table 3). This improvement in predictive accuracy, along with the ability of the SHAP framework to reveal nonlinear feature interactions, highlights the clear advantage and added value of the applied machine learning strategy.

3.3 Survival analysis based on risk stratification

Survival outcomes stratified by the model-predicted risk are shown in Supplementary Fig. 2 (Kaplan–Meier curves). The log-rank test confirmed a statistically significant difference in cardiovascular mortality between high-risk and low-risk groups,

Table 1 Baseline characteristics of the participants

Variable(*n*)	Level	Survivor (*n* = 3366)	Cardiac mortality (*n* = 256)	Overall (*n* = 3622)	*P* value
Gender (%)	Female	1546 (45.9%)	156 (60.9%)	1702 (47.0%)	< 0.001
	Male	1820 (54.1%)	100 (39.1%)	1920 (53.0%)	
Age	Mean (SD)	64.8 (14.3)	76.4 (7.2)	65.6 (14.2)	< 0.001
Race (%)	Mexican American	240 (7.1%)	4 (1.6%)	244 (6.7%)	
	Other Hispanic	217 (6.4%)	4 (1.6%)	221 (6.1%)	
	Non-Hispanic White	2209 (65.6%)	212 (82.8%)	2421 (66.8%)	
	Non-Hispanic Black	507 (15.1%)	34 (13.3%)	541 (14.9%)	
	Other Race	193 (5.7%)	2 (0.8%)	195 (5.4%)	
Education (%)	< High school	310 (9.2%)	32 (12.5%)	342 (9.4%)	0.035
	Completed high school	416 (12.4%)	41 (16.0%)	457 (12.6%)	
	> High school	2640 (78.4%)	183 (71.5%)	2823 (77.9%)	
Marital (%)	Married/living with partner	2015 (59.9%)	132 (51.6%)	2147 (59.3%)	0.011
	Widowed/divorced/separated/never married	1351 (40.1%)	124 (48.4%)	1475 (40.7%)	
Smoke (%)	Never	1546 (45.9%)	104 (40.6%)	1650 (45.6%)	< 0.001
	Former	1269 (37.7%)	127 (49.6%)	1396 (38.5%)	
	Current	551 (16.4%)	25 (9.8%)	576 (15.9%)	
PIR (%)	Low income	813 (24.2%)	62 (24.2%)	875 (24.2%)	< 0.001
	Middle income	1431 (42.5%)	141 (55.1%)	1572 (43.4%)	
	High income	1122 (33.3%)	53 (20.7%)	1175 (32.4%)	
Diabetes (%)	Yes	2702 (80.3%)	181 (70.7%)	2883 (79.6%)	< 0.001
	No	664 (19.7%)	75 (29.3%)	739 (20.4%)	
Alcohol (%)	Yes	1312 (39%)	94 (36.7%)	1406 (38.8%)	0.517
	No	2054 (61.0%)	162 (63.3%)	2216 (61.2%)	
Heart failure (%)	Yes	3134 (93.1%)	206 (80.5%)	3340 (92.2%)	< 0.001
	No	232 (6.9%)	50 (19.5%)	282 (7.8%)	
Coronary heart disease (%)	Yes	3061 (90.9%)	192 (75.0%)	3253 (89.8%)	< 0.001
	No	305 (9.1%)	64 (25.0%)	369 (10.2%)	
Angina pectoris (%)	Yes	3178 (94.4%)	217 (84.8%)	3395 (93.7%)	< 0.001
	No	188 (5.6%)	39 (15.2%)	227 (6.3%)	
Heart_disease (%)	Yes	3058 (90.8%)	197 (77.0%)	3255 (89.9%)	< 0.001
	No	308 (9.2%)	59 (23.0%)	367 (10.1%)	
Stroke (%)	Yes	3100 (92.1%)	201 (78.5%)	3301 (91.1%)	< 0.001
	No	266 (7.9%)	55 (21.5%)	321 (8.9%)	
Hypercholesterolemia (%)	Yes	1610 (47.8%)	112 (43.8%)	1722 (47.5%)	0.232
	No	1756 (52.2%)	144 (56.2%)	1900 (52.5%)	
Hypertension (%)	Yes	1466 (43.6%)	67 (26.2%)	1533 (42.3%)	< 0.001
	No	1900 (56.4%)	189 (73.8%)	2089 (57.7%)	
BMI (%)	No obesity	1660 (49.3%)	133 (52%)	1793 (49.5%)	0.454
	Obesity	1706 (50.7%)	123 (48.0%)	1829 (50.5%)	
Depression (%)	No depression	2351 (69.8%)	169 (66.0%)	2520 (69.6%)	0.014
	Mild depression	660 (19.6%)	68 (26.6%)	728 (20.1%)	
	Moderate to severe depression	355 (10.5%)	19 (7.4%)	374 (10.3%)	
Sleep disorder (%)	No sleep disorder	1884 (56%)	147 (57.4%)	2031 (56.1%)	0.700
	Sleep disorder	1482 (44.0%)	109 (42.6%)	1591 (43.9%)	
Log_NLR	Mean (SD)	0.8 (0.5, 1.1)	1 (0.7, 1.4)	0.8 (0.5, 1.1)	< 0.001
CRP(mg/l)	Mean (SD)	3.3 (1.5, 5.9)	4.3 (1.4, 9)	3.3 (1.5, 6.1)	< 0.001

Table 2 AUC values of the five machine algorithms

	AUC of the best combination of parameters in the training set	AUC of the best combination of parameters in the validation set
RSF	0.989 (0.985–0.992)	0.831 (0.779–0.879)
GBM	0.927 (0.890–0.956)	0.935 (0.905–0.960)
Lasso + Cox	0.819 (0.762–0.879)	0.849 (0.792–0.895)
Coxboost	0.818 (0.763–0.874)	0.85 (0.794–0.898)
SVM	0.833 (0.778–0.878)	0.832 (0.778–0.879)

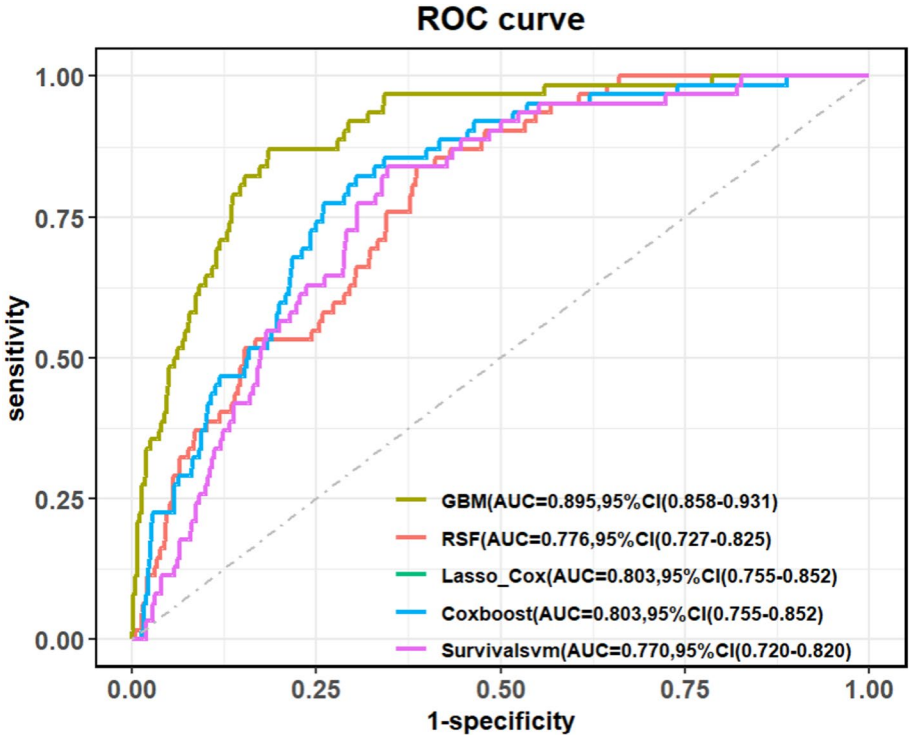


Fig. 1 ROC curve analysis of five machine learning algorithms for predicting cardiovascular mortality in cancer survivors in the validation set

Table 3 C-index of the five machine algorithms

	The C-index of the best combination of parameters in the training set	The C-index of the best combination of parameters in the validation set
RSF	0.986 (0.982–0.989)	0.782 (0.723–0.837)
GBM	0.919 (0.891–0.944)	0.914 (0.868–0.949)
Lasso + Cox	0.835 (0.790–0.875)	0.821 (0.766–0.874)
Coxboost	0.835 (0.794–0.876)	0.822 (0.768–0.869)
SVM	0.818 (0.780–0.853)	0.795 (0.738–0.845)

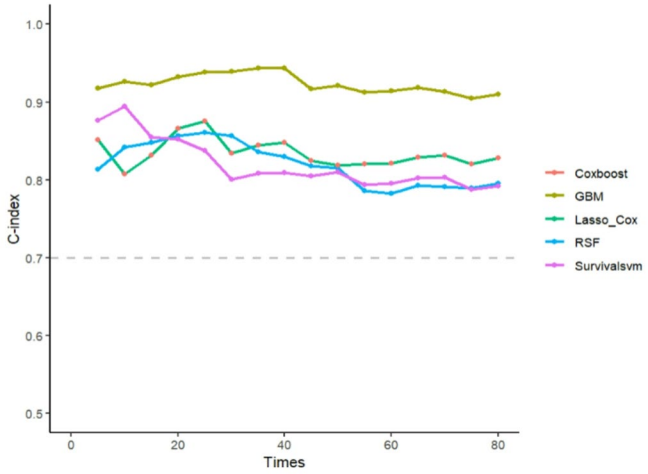


Fig. 2 Time-dependent C-index curves of five machine learning algorithms for predicting cardiovascular mortality among cancer survivors in the validation set

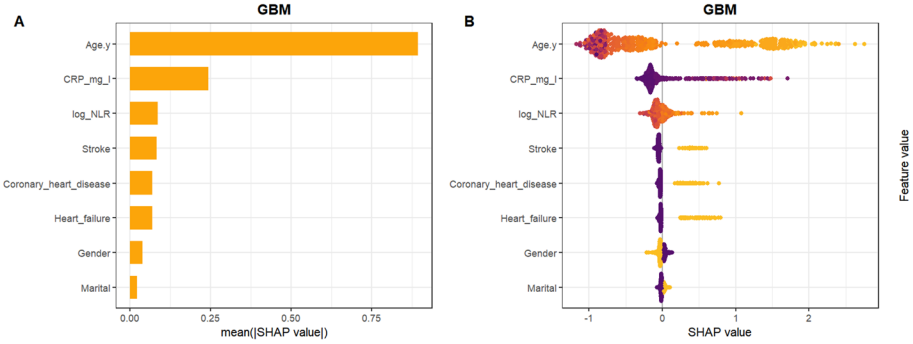


Fig. 3 **A** The eight variables are ranked according to the importance of the mean value; **B** ranking of the importance of the 8 risk factors for stability and interpretation using the optimal model. *CRP* C-reactive protein, *NLR* neutrophil/lymphocyte ratio

dichotomized by the median risk score. The high-risk group showed significantly worse survival ($P < 0.001$).

3.4 Interpretability analysis using SHAP

3.4.1 Global feature importance of the GBM model

We used the SHAP framework to quantify and visualize each predictor's contribution. Feature importance, based on mean absolute SHAP values, is ranked in Fig. 3A. Figure 3B summarizes these eight features, illustrating the distribution and direction (risk-increasing or risk-decreasing) of their SHAP values across patients. The top five predictors were age, CRP, NLR, stroke, and coronary heart disease.

3.4.2 Interaction effects involving age

Considering the model's identification of age as a dominant feature, we specifically explored its interactions with other key predictors: sex, marital status, heart failure, coronary heart disease, stroke, NLR, and CRP. The SHAP dependence plots (Fig. 4) reveal that the predictive contributions of a history of stroke, coronary heart disease, or heart

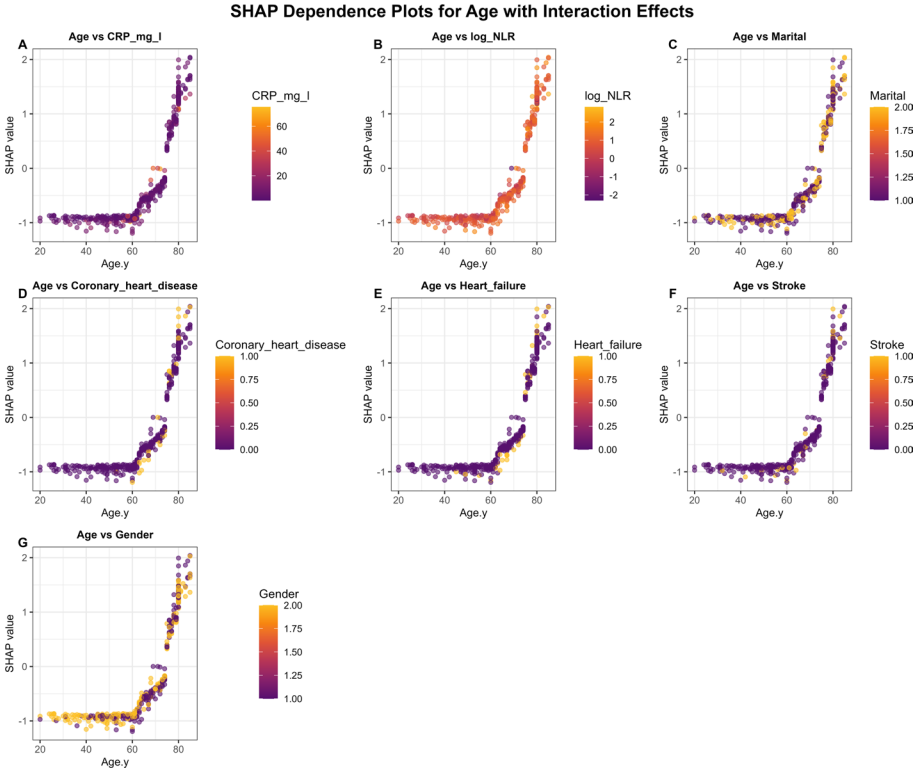


Fig. 4 Interaction between dependence plots and age. The figure illustrates the interaction between characteristics and age. **A** CRP, **B** NLR, **C** stroke, **D** coronary heart disease, **E** heart failure, **F** gender G: marriage

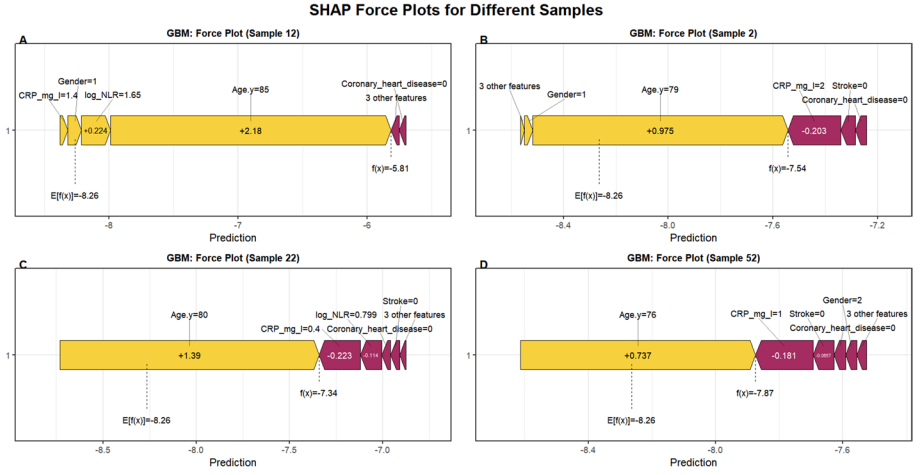


Fig. 5 Distribution of SHAP values in the individual prediction analysis (**A**–**D**) of the four patients

failure were substantially amplified in older patients. Conversely, the influences of sex, marital status, CRP level, and NLR did not exhibit consistent trends across age groups.

3.4.3 Local interpretability: case-based SHAP analysis

To illustrate interpretability at an individual level, we performed local SHAP analyses for four representative patients. For each patient, the model generated a prediction, and SHAP analysis decomposed the contribution of each predictor to the patient's risk score (Fig. 5A–D). This case-based analysis visualizes how the importance and direction

(risk-increasing or risk-decreasing) of key factors, such as age, CRP level, or comorbidity history, differ substantially among individuals.

3.5 Development of a clinical prediction tool (nomogram)

The predictor selection process for the clinical nomogram is summarized below. First, LASSO regression was applied to an initial set of covariates, and 14 variables were selected for inclusion in a Cox proportional hazards model. From this model, eight variables remained statistically significant independent predictors: age, sex, marital status, histories of heart failure, coronary heart disease, and stroke, as well as NLR and CRP levels. These eight predictors were incorporated into a nomogram to estimate the probability of cardiovascular death at specific time points (Supplementary Fig. 3). The nomogram showed good discriminative performance, with an AUC of 0.868 (Supplementary Fig. 4). Finally, decision curve analysis (DCA) confirmed the clinical utility of the nomogram by demonstrating a positive net benefit across a wide range of threshold probabilities (Supplementary Fig. 5).

3.6 Sensitivity analysis

To evaluate the robustness of the model and address concerns that predictions might be driven primarily by established CVD, a sensitivity analysis was performed. All participants with a baseline history of heart failure or coronary heart disease were excluded. The performance of the optimal GBM model was then reassessed in this restricted subgroup. In the validation set, the model achieved an AUC of 0.955 (95% CI 0.879–0.988) and a C-index of 0.901 (95% CI 0.867–0.941). These findings confirm that the model maintains strong discriminative ability for stratifying cardiovascular death risk, even among cancer survivors without pre-existing cardiac conditions.

3.7 Deployment and accessibility of the web-based prediction tool

The web-based tool "CardioRisk-CA" has been developed and deployed based on the optimal GBM model. The tool is publicly available and is hosted at the following stable URL: https://xiaoheshi.shinyapps.io/cardiac-app/.

3.8 External validation in the SEER database

External validation was performed using the SEER registry (2010–2021). This assessed the model's generalizability to a large independent clinical cohort. The validation cohort included 970,777 cancer patients with a median follow-up of 5.83 years, during which 46,677 cardiovascular deaths occurred (baseline characteristics in Supplementary Table 4). The GBM model showed good discriminative performance in this independent cohort, achieving a C-index of 0.815 and an AUC of 0.866 (Supplementary Table 5). Global interpretation using SHAP analysis confirmed age as the primary risk factor. Primary cancer site and treatment modality were the next most influential features. In contrast, sex, cancer stage, and marital status exhibited weaker impacts. Comprehensive feature importance rankings and SHAP distributions are visualized in Supplementary Fig. 6.

4 Discussion

This study systematically established and assessed five machine learning-based survival prediction algorithms (GBM, RSF, CoxBoost, LASSO + Cox, and Survival SVM) using prospective cohort data derived from the NHANES to estimate cardiovascular mortality risk among cancer survivors. Among the tested models, the GBM exhibited the highest predictive accuracy in the validation cohort (AUC = 0.935, C-index = 0.914). Age, marital status, sex, CRP, NLR, and histories of stroke, coronary heart disease, and heart failure were identified as essential predictors. Furthermore, using the SHAP interpretability approach, we elucidated both global and individual contributions of these variables, as well as interaction effects involving age with stroke, coronary heart disease, and heart failure. Additionally, the nomogram constructed by employing LASSO regression for predictor selection demonstrated robust clinical applicability (AUC = 0.868). To further advance the clinical translation of the model, we deployed the best-performing GBM model as an interactive web-based tool, CardioRisk-CA (Cardiovascular Risk Calculator for Cancer Survivors). This tool enables convenient, personalized risk assessment with visualized explanations for clinicians and provides an open platform for external validation and multicenter collaboration.

Moreover, a comprehensive external validation of the GBM model was conducted utilizing the SEER database from the National Cancer Institute to examine its performance beyond the NHANES dataset. In this independent cohort, the GBM model retained excellent discrimination capability (C-index = 0.815, AUC = 0.866), confirming strong model generalizability. SHAP analysis performed on SEER data consistently revealed age as the dominant predictor of cardiovascular mortality, while also highlighting the substantial predictive relevance of cancer-specific factors, such as primary tumor site and treatment methods. Conversely, variables such as sex, marital status, and cancer stage demonstrated relatively limited influence. These findings imply that despite lacking detailed laboratory measurements such as NLR and CRP in typical clinical databases, the model still effectively stratifies cardiovascular mortality risk by incorporating easily accessible demographic information, cancer-specific attributes, and prevalent comorbid conditions, thus increasing its practicality and applicability in clinical scenarios.

Age was identified as the most significant predictor in this study, aligning with the established biological understanding that aging is a fundamental risk factor for multiple chronic diseases [17]. Aging involves a gradual deterioration of physiological functions at the cellular and systemic levels [18]. Its strong relationship with CVD and cancer is evident through specific biomarkers such as CRP [19–21]. Thus, age significantly influences cardiovascular mortality risk in cancer survivors.

Inflammation represents a common pathological basis underlying both CVD and cancer [22]. Chronic inflammation is not only instrumental in multiple stages of tumor initiation and progression but also central to the pathogenesis of atherosclerosis [23]. CRP, an established biomarker of systemic inflammation, is strongly associated with elevated CVD risk [24]. It contributes to atherogenesis via several mechanisms, including complement activation, endothelial dysfunction, and plaque destabilization [25]. For instance, CRP disrupts endothelium-dependent vasodilation by suppressing nitric oxide release and stimulating endothelin-1 production, consequently causing endothelial impairment [26, 27]. Additionally, CRP promotes expression of matrix metalloproteinases (MMP-1, MMP-2, and MMP-9), which facilitate plaque instability [28, 29]. Another inflammatory

marker, the NLR, has been linked to poor prognosis in atherosclerotic CVDs [30, 31]. Our identification of both CRP and NLR as essential predictors of cardiovascular death among cancer survivors underscores the critical role of inflammation in this scenario. Moreover, the low-grade chronic inflammation associated with aging may further aggravate cardiovascular risk [32].

Multiple risk factors, including age, smoking, and obesity, are common to both CVD and cancer. Their intertwined pathophysiological mechanisms involve several biological pathways such as inflammation, neurohormonal activation, circulating mediators, immune dysregulation, and oxidative stress [33]. Inflammation, a central among these mechanisms, was highlighted in the CANTOS trial, where inhibition of IL-1β notably reduced coronary event risk [34]. Additionally, heart failure may accelerate tumor growth through secretion of cardiokines [35]. Chronic activation of the renin-angiotensin-aldosterone system (RAAS), although initially beneficial to cardiac function, can lead to cumulative cardiorenal damage over time [36]. Mitochondrial oxidative phosphorylation, critical in cardiac energy metabolism, also significantly contributes to cancer progression [37]. Certain chemotherapy drugs (e.g., fluoropyrimidines) can directly trigger myocardial ischemia through mechanisms such as coronary artery vasospasm [38]. These shared risk factors and interconnected pathological pathways result in a reciprocal exacerbation between CVD and cancer, significantly increasing cardiovascular mortality risk among cancer survivors.

Stroke and cancer share common risk factors; however, the pathogenesis of stroke in cancer patients exhibits distinct characteristics. Tumors can cause stroke by directly compressing or invading blood vessels or by creating a hypercoagulable state that predisposes patients to embolic events [39–43]. Cancer-related hypercoagulability involves various mechanisms, including inflammatory mediator release from tumor cells, promotion of platelet aggregation, and activation of coagulation pathways [44]. Furthermore, cancer patients exhibit an increased prevalence of patent foramen ovale (PFO) and venous thromboembolism (VTE), further elevating stroke risk through paradoxical embolism [40]. Population-based studies have demonstrated that cancer patients experience significantly higher stroke-related mortality [standardized mortality ratio (SMR) = 2.17] [45]. Consistent with these findings, our study identifies prior stroke history, similar to established CVD, as a significant predictor of cardiovascular death among cancer survivors.

Research indicates that among young adult cancer survivors, women have a higher risk of CVD compared with non-cancer controls, while no significant difference exists between male survivors and their counterparts [46]. This pattern contrasts with the general population, where women usually exhibit lower cardiovascular risk compared with men. Such discrepancies may stem from cardiotoxicity related to cancer treatment (anthracyclines, for instance, exert greater cardiotoxic effects in women [47, 48]) and sex-specific physiological responses, including higher short-term mortality in women after acute myocardial infarction [49, 50]. Additionally, genetic factors such as the presence of the ApoE4 allele significantly elevate mortality risk in middle-aged and older women, with a less pronounced effect in men [51]. Thus, female sex emerges as an essential predictor of cardiovascular mortality risk among cancer survivors.

Marital status is likewise associated with cardiovascular outcomes in this patient population. Previous research demonstrates that being single, separated, divorced, or

widowed correlates with worse cardiovascular outcomes [52, 53]. This relationship may be attributable to differences in available social support. Married or cohabiting individuals typically benefit from improved physical care and psychosocial support, thereby enhancing treatment adherence and overall health, and consequently reducing cardiovascular mortality risk [53].

The external validation performed using the SEER database further underscores the significant role of cancer-specific characteristics in cardiovascular mortality risk. These findings align with existing knowledge in cardio-oncology, where distinct cancer types possess markedly different cardiovascular risk profiles due to biological variability, treatment approaches, and systemic effects. For instance, breast cancer survivors are notably susceptible to myocardial injury and heart failure due to treatments such as anthracyclines, trastuzumab, and thoracic radiotherapy [54, 55]. Furthermore, combined therapeutic approaches, such as chemotherapy plus radiotherapy, may induce synergistic cardiotoxic effects. By integrating these factors, our model provides population-level quantification of the cardiovascular risk associated with different cancer types and treatment regimens. This approach offers a novel, data-driven foundation for refining cancer-specific cardiovascular risk management strategies in clinical practice. Future model improvements could incorporate more detailed information on treatment dosage and duration to further enhance personalized cardiovascular risk stratification.

Our methodological pathway, integrating robust feature selection (consistent with the framework proposed by Wang et al. [10]), high-performance machine learning, explainable AI, and deployment as an accessible web-based tool—offers a complete pipeline from "data" to "decision" for developing similar clinical prediction systems, demonstrating strong translational value. Partial external validation using the large-scale, independent SEER dataset effectively addressed limitations arising from the NHANES database's lack of cancer-specific information (e.g., cancer type, stage, and treatment details) and demonstrated the robust generalizability of the proposed model. The primary contribution of this study is the translation of machine learning predictions into a clinically actionable decision-support tool. The online platform encapsulates the predictive algorithm and integrates SHAP-based interpretability, thereby transforming a "black-box" model into a transparent and actionable system. In clinical practice, this tool can be applied in several known scenarios. First, in oncology or cardiology outpatient clinics, physicians can use a streamlined input interface to rapidly assess long-term cardiovascular risk in cancer survivors, enabling early risk stratification and prioritized intervention. Second, case-specific SHAP explanations (e.g., the age–stroke history interaction) can support doctor–patient communication and assist in developing individualized monitoring and prevention strategies, thereby improving patient adherence and self-management. Furthermore, the tool's built-in batch validation function supports external validation across diverse populations and institutions, facilitating assessment of generalizability and promoting the development of a broader cardiovascular risk registry for cancer survivors.

However, this study has limitations, including reliance on static baseline data and the inability to capture dynamic risk evolution. Future studies may incorporate multisource longitudinal data, such as electronic health records and time-varying biomarkers, and apply time-series models to further improve timeliness and personalization of

predictions. Ultimately, these advances may enhance the precision and proactivity of cardiovascular risk management in cancer survivors.

5 Conclusion

This study developed and validated an interpretable machine learning model, enhanced by SHAP, for predicting cardiovascular mortality risk among American cancer survivors. Deployed as an accessible web-based tool (CardioRisk-CA), it offers a practical, data-driven approach to risk assessment and establishes a platform for both personalized risk management and multicenter validation, thereby serving both clinical and research purposes.

Supplementary Information

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Supplementary Material 1.

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Author contributions
Xiaohe Shi: contributed to the design of the study protocol and wrote the first draft. Yinliang Liu: participated in the extraction and analysis of the data and reviewed and edited it. All the authors contributed to the manuscript and approved the version for submission. All the authors read and approved the final manuscript.

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Data availability
The datasets analyzed during the current study are publicly available from the NHANES repository: http://www.cdc.gov/nchs/nhanes.htm. The SEER data are available from the National Cancer Institute's SEER Program (https://seer.cancer.gov/). The interactive web-based tool "CardioRisk-CA" developed in this study is publicly available at: https://xiaoheshi.shinyapps.io/cardiac-app/.

Declarations

Ethics approval and consent to participate
The use of NHANES data in this study was approved by the Ethics Review Board of the National Center for Health Statistics (NCHS). SEER data constitute publicly available, de-identified information, and its use complies with relevant U.S. laws and regulations. All NHANES participants provided written informed consent prior to the survey, agreeing to the use of their data for scientific research.

Consent for publication
All authors have consented to the publication of this manuscript.

Clinical trial number
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The authors declare no competing interests.

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